Project Title:
Identifying High-Potential but Under-Engaged Customers Through Education, Income, Campaign Behavior, and Macroeconomic Context

Research Question

Although higher education is associated with greater spending, campaign engagement does not scale uniformly across education tiers — and the highest concentration of high-spending but non-responsive customers is found among lower education groups. This project investigates whether education level, relative income, and macroeconomic context at enrollment (CPI) can jointly explain campaign disengagement and identify the most untapped customer segments.

This project investigates a behavioral pattern observed in customer data: spending increases with education level, yet a substantial portion of high-spending customers across all education tiers has never responded to any of the six marketing campaigns in the dataset. Notably, the highest share of high-potential but under-engaged customers is observed among lower education groups rather than higher ones. The central question is: Can education level, relative income positioning, campaign response history, and the macroeconomic environment at enrollment (CPI) jointly identify these under-engaged segments — and what distinguishes customers who respond to campaigns from similarly situated customers who do not?

The primary dataset is the Customer Personality Analysis dataset (2,240 customers, 29 features), covering enrollments from January to December 2013. It includes demographic variables (education across 5 tiers, marital status, household composition), product-level spending (wines, fruits, meat, fish, sweets, gold), purchase channel data, and six binary campaign response indicators (AcceptedCmp1-5 and Response), which will be combined into a composite Campaign Engagement Score. This is enriched with monthly US CPI data (918 observations, 1947-2023), matched to each customer's enrollment month to capture inflation context, and an external salary benchmark by education level to construct a relative income measure.

The analysis will include data preprocessing, RFM feature engineering, CPI cohort tagging, and relative income construction, followed by statistical hypothesis testing across education groups. The expected outcome is a data-driven characterization of high-potential but under-engaged customer segments and actionable targeting recommendations.